

## Mark scheme – Level 3 Certificate Mathematical Studies

### Paper 2A Statistical Techniques – Predicted June 2026 (Computer Games Market)

Total: 60 marks

#### Question 1 (6 marks)

Use Report A.

##### 1(a) [1 mark]

**Answer:** Any one valid reason, e.g.

- **B1:** Forecast growth 2023F–2026F is described as “more conservative” / lower than before 2020 / significantly lower than 2018–2021.
- Or: flat growth trend 2021–2023F suggests slower future growth.
- Or: market already large/mature so percentage growth likely to slow.

(Any one clearly linked to Report A.)

##### 1(b) [2 marks]

2023F revenue = \$94.0 bn Increase = \$13.4 bn

Percentage increase =  $\frac{13.4}{94.0} \times 100 \approx 14.3\%$

- **M1:** Uses  $\frac{\text{increase}}{\text{original}} \times 100$  with 13.4 and 94.0.
- **A1:** 14.3% (accept 14%–14.4%).

##### 1(c) [3 marks]

Key ideas:

- Revenue depends on **spend per player**, not just number of players.
- PC and console may have **different average spend** or pricing models.
- Starting numbers of players differ, so same CAGR in players does not imply same revenue growth.

Award, for example:

- **B1:** States that revenue depends on both number of players and how much each player spends.
- **B1:** Notes that console and PC may have different average spend / monetisation models.
- **B1:** Concludes that a higher player CAGR for consoles does not guarantee faster revenue growth.

(Any three clear, non-repetitive points.)

## Question 2 (10 marks)

Use Graph 1.

### 2(a) [3 marks]

From Graph 1:

- Total revenue 2021  $\approx$  \$94.3 bn
- Total revenue 2023F  $\approx$  \$94.0 bn

Change is very small (about \$0.3 bn).

- **B1:** Identifies that PC/total revenue changes only slightly between 2021 and 2023F (about \$0.3 bn).
- **B1:** States that this is not a “sharp” fall (describes as small / almost flat).
- **B1:** Concludes Journalist A’s claim is not valid / exaggerated.

### 2(b) [4 marks]

From Graph 1:

- 2022–2023 growth:  $-\$0.3$  bn (or a fall of \$0.3 bn).
- 2024–2026 growth:  $+\$13.4$  bn.

Compare:

- $|-0.3|$  vs 13.4.
- 13.4 is far more than double 0.3.

Marking:

- **M1:** Identifies 2022–2023 change as about  $-\$0.3$  bn (or fall of \$0.3 bn).
- **M1:** Identifies 2024–2026 change as about  $+\$13.4$  bn.
- **M1:** Compares the two (e.g. 13.4 is much larger than 0.3).
- **A1:** States that the claim “more than double” is technically true but misleading because one period is a small fall and the other a large rise, or clearly concludes appropriately.

(Allow full credit if they treat 0.3 as a small positive “growth” but still correctly conclude 13.4 is far more than double.)

### 2(c) [1 mark]

- **B1:** States that comparing periods of different lengths (e.g. 1 year vs 3 years) is misleading unless adjusted to per-year rates / same time length.

### 2(d) [2 marks]

Any two sensible improvements, e.g.

- **B1:** Add exact revenue values on the bars / a clear scale.
- **B1:** Separate PC and console bars more clearly or use consistent shading / legend; or show growth per year rather than total over different periods.

(Any two distinct, reasonable suggestions.)

## Question 3 (6 marks)

Use Graph 2.

### 3(a) [2 marks]

From Graph 2:

- PC: Premium 56%, Microtransactions 30%, Downloads 12%, Subscription 2%.
- Console: Premium 57%, Microtransactions 33%, Downloads 8%, Subscription 2%.

Any two differences, e.g.

- **B1:** PC has a higher percentage from downloads (12% vs 8%).
- **B1:** Consoles have a higher percentage from microtransactions (33% vs 30%).

(Allow other correct comparisons; premium and subscription are almost the same, so must be stated accurately.)

### 3(b) [2 marks]

Premium share = 56% of \$62.2 bn.

$$0.56 \times 62.2 = 34.832 \text{ bn}$$

- **M1:** Attempts  $0.56 \times 62.2$ .
- **A1:** \$34.8 bn (accept \$34.83 bn or \$34.8 billion).

### 3(c) [2 marks]

Key ideas:

- More time spent in games → more opportunities for microtransactions.
- Even if premium sales stay strong, players can continue to spend repeatedly in-game.

Award, for example:

- **B1:** States that microtransactions are recurring / can happen many times per player.
- **B1:** Explains that as players spend more time in games, in-game spending can grow as a share of revenue even if premium sales remain high.

## Question 4 (8 marks)

Use Report B.

### 4(a) [2 marks]

Reasons survey may not represent all players globally, e.g.

- Sample may be from one region / platform / age group only.
- Sample size or method may be biased (online survey, self-selection).
- **B1:** Identifies a possible lack of representativeness (e.g. only certain countries / platforms).
- **B1:** Explains why that makes it unrepresentative of all global players.

### 4(b) [2 marks]

Survey result: 53% prefer F2P.

Claim: “Fewer than half prefer F2P.”

- **M1:** Uses 53% from the survey.
- **A1:** Concludes the claim is not supported / false because  $53\% > 50\%$ .

### 4(c) [1 mark]

Subscription preference = 9%.

$$0.09 \times 80 = 7.2$$

- **B1:** 7 (or 7.2 with clear rounding to 7).

### 4(d) [3 marks]

Key ideas:

- Larger sample reduces random variation / sampling error.
- Estimates become more stable / closer to true population proportion.

Award, for example:

- **B1:** States that a larger sample reduces the effect of anomalies / random variation.
- **B1:** States that the estimate is more likely to be close to the true population value.
- **B1:** Clear link to reliability (e.g. “results are more reliable / more representative”).

## Question 5 (7 marks)

### 5(a) [3 marks]

Class widths and frequency densities:

- 0–5: width 5  $\rightarrow 12/5=2.4$
- 5–10: width 5  $\rightarrow 18/5=3.6$
- 10–20: width 10  $\rightarrow 20/10=2.0$
- 20–30: width 10  $\rightarrow 7/10=0.7$
- 30–50: width 20  $\rightarrow 3/20=0.15$

Marking:

- **M1:** Uses frequency density = frequency  $\div$  class width.
- **B1:** At least three correct densities.
- **B1:** All five correct (allow rounding to 0.15).

### 5(b) [4 marks]

Histogram:

- Correct class boundaries on horizontal axis: 0, 5, 10, 20, 30, 50.
- Bars with correct widths.
- Heights proportional to densities: 2.4, 3.6, 2.0, 0.7, 0.15.

Marking:

- **B1:** Correct class intervals on the axis.
- **B1:** Bars drawn with correct widths in correct positions.
- **M1:** Heights consistent with their densities from 5(a).
- **A1:** Fully correct histogram (all heights and positions correct).

(If 5(a) wrong but 5(b) consistent with their densities, allow method marks.)

## Question 6 (8 marks)

### 6(a) [2 marks]

From the table, as hours increase, spending increases roughly linearly.

- **B1:** Positive correlation.
- **B1:** Strong (or fairly strong) correlation.

### 6(b) [3 marks]

Given data:

$$\sum x=74, \sum y=113, \sum x^2=830, \sum xy=1279, n=8$$

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2} = \frac{8 \times 1279 - 74 \times 113}{8 \times 830 - 74^2} = \frac{1870}{1164} \approx 1.61$$

$$\bar{x} = 9.25, \bar{y} = 14.125$$

$$a = \bar{y} - b\bar{x} \approx 14.125 - 1.61 \times 9.25 \approx -0.73$$

Regression line:

$$y \approx -0.73 + 1.61x$$

Marking:

- **M1:** Correct formula for b with substitution.
- **A1:**  $b \approx 1.61$  (3 s.f.).
- **A1:**  $a \approx -0.73$  (accept  $-0.73$  to  $-0.733$  s.f.) and equation  $y = a + bx$ .

(Allow equivalent rounded values.)

### 6(c) [1 mark]

Use regression line:

$$y \approx -0.73 + 1.61 \times 11 \approx -0.73 + 17.71 \approx 16.98$$

- **B1:** Approximate answer \$17.0 (accept 16.9–17.0).

### 6(d) [2 marks]

Key idea: extrapolation.

- **B1:** States that 40 hours is far outside the range of the data (3–16 hours).
- **B1:** Explains that the linear relationship may not hold for such large values, so the estimate may be unreliable.

## Question 7 (8 marks)

Weekly spending on cloud gaming subscriptions:  $X \sim N(18.5, 6.22)$ .

### 7(a) [1 mark]

- **B1:**  $X \sim N(18.5, 6.22)$ .

### 7(b) [3 marks]

$$P(X > 25) = 1 - P(Z \leq 25 - 18.56.2) \\ z = 6.56.2 \approx 1.05$$

From tables,  $\Phi(1.05) \approx 0.853$ .

$$P(X > 25) \approx 1 - 0.853 = 0.147$$

- **M1:** Standardises correctly:  $z = 25 - 18.56.2$ .
- **M1:** Uses normal table to find  $\Phi(z)$ .
- **A1:** Probability  $\approx 0.15$  (accept 0.14–0.15).

### 7(c) [3 marks]

$$P(12 < X < 30) = \Phi(30 - 18.56.2) - \Phi(12 - 18.56.2)$$

$$z_1 = 30 - 18.56.2 = 11.56.2 \approx 1.85$$

$$z_2 = 12 - 18.56.2 = -6.56.2 \approx -1.05$$

From tables:

$$\Phi(1.85) \approx 0.968, \Phi(-1.05) \approx 1 - 0.853 = 0.147.$$

$$P(12 < X < 30) \approx 0.968 - 0.147 = 0.821$$

- **M1:** Standardises both bounds correctly.
- **M1:** Uses normal tables for both z-values.
- **A1:** Probability  $\approx 0.82$  (accept 0.81–0.83).

### 7(d) [1 mark]

- **B1:** Explains that the normal distribution is continuous and extends below 0, whereas spending cannot be negative and many players may spend exactly \$0, so the model is poor at that end.

## Question 8 (7 marks)

Sample:  $n=40$ ,  $\bar{x}=14.20$ ,  $\sigma=5.80$ . Assume population standard deviation known.

### 8(a) [4 marks]

95% confidence interval:

$$\bar{x} \pm z_{0.975} \cdot \sigma/n$$

$$z_{0.975} = 1.96, \sigma/n = 5.840 \approx 5.86.3249 \approx 0.917$$

$$\text{Margin} = 1.96 \times 0.917 \approx 1.80$$

$$CI \approx 14.20 \pm 1.80 \Rightarrow (12.4, 16.0)$$

Marking:

- **M1:** Uses correct CI structure with z and  $\sigma/n$ .
- **M1:** Correctly computes  $\sigma/n$ .
- **M1:** Multiplies by 1.96 to get margin.
- **A1:** Interval approximately (12.4, 16.0) (accept small rounding differences).

### 8(b) [3 marks]

Claim: “Average player spends at least \$16 per week.”

From CI:

- Upper limit  $\approx \$16.0$ , lower limit  $\approx \$12.4$ .

Interpretation:

- **B1:** Notes that 16 is at/near the upper end of the interval and values below 16 are plausible.
- **B1:** States that the interval suggests the true mean could be less than \$16.
- **B1:** Concludes that the claim is not supported / cannot be confidently accepted based on this sample.